

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 24/05/2018

Day: Thursday

Time: 10:00 am to 10:30 am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : BOP- May 2018
Semester: 1
Section : I
Course Code : BOP 104
Course Name: Geometrical Optics-I

Important Instructions:

- There are two sections in the paper. **SECTION –I** contain 20 Questions and **SECTION – II** contains 3 Questions.
- SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs)
(Encircle the correct answer)

(10 × 1 = 10)

- Interference phenomenon of waves was given by.....**
 - Isaac Newton
 - Christian Huygen
 - Thomas Young
 - James Clerk Maxwell
- The range of the visible light in the electromagnetic spectrum is between.....**
 - 0-200 nm
 - 200-400 nm
 - 400-700 nm
 - 700-1000 nm
- Light travel faster in**
 - Water
 - Diamond
 - Glass
 - Plastic

4. When wave enter from one medium to another medium doesn't change.
 - a. Frequency
 - b. Velocity
 - c. Wavelength
 - d. All of above
5. In case of mirror, reflection angle becomes..... that of angle of rotation.
 - a. Half
 - b. Twice
 - c. Thrice
 - d. No change
6. Spilling of white light into different color is called.....
 - a. Diffraction
 - b. Interference
 - c. Polarization
 - d. Dispersion
7. Lens with one face concave and other concave is called
 - a. Converging lens
 - b. Diverging lens
 - c. Miniscus lens
 - d. Plano lens
8. Refracted angle is larger for..... color light.
 - a. Violet
 - b. Red
 - c. Green
 - d. Yellow
9. Power of the lens is given by.....
 - a. f/μ
 - b. $1/f$
 - c. μ/f
 - d. $1/\mu$
10. For divergence, equation of vergence is given by.....
 - a. n/r
 - b. r/n
 - c. $-n/r$
 - d. $-r/n$

Identify the sentence as True or False:

(Please tick mark the answer)

11. Midpoint of a spherical mirror is called focal point.
12. Speed of light in vacuum is 3×10^8 cm/s.
13. Convex lens is use for convergence.
14. Bending of light is more in diamond compare to glass.
15. Relation between velocity, wavelength and frequency is given by,
 $\lambda = Vf$

(5 × 1 = 5)	
True	False
☐	☐
☐	☐
☐	☐
☐	☐
☐	☐

P.T.O

(5 × 1 = 5)

Match the following

16. Total internal reflection		A. Sir Isaac Newton
17. Particle Nature		B. Constant
18. Vergence		C. Critical angle
19. Prism		D. Curvature of wavefront
20. Refractive index		E. Diopter

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st SEM B.O.P Examination

May 2018

BOP 104 Geometrical Optics-I

Date: 24/05/2018

Time: 10:30 am to 01:00 pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (10 out of 11) (Answer not exceeding 50 words) (10 × 2 = 20)

- a. Define wavelength and frequency in terms of sine wave.
- b. Why speed of light decreases when it travels through the medium?
- c. Prove that refractive index of the medium depends upon the wavelength of the light.
- d. Define absolute and relative refractive index.
- e. Define focal length and radius of curvature.
- f. Mention sign convention for convex and concave mirror.
- g. Define prism and its types.
- h. Define types of lens with lens maker equation.
- i. Define prism diopter with equation.
- j. Define lateral and angular magnification.
- k. Define principle plane and principle point for lens.

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Write a short note on wavefront.
- b. Derive the law of reflection and law of refraction by Fermat's principle.
- c. Derive the equation for angle of deviation $\delta = A(\mu - 1)$.
- d. Write a short note on nodal point and nodal plane. (with proper diagram)
- e. Write a short note on crown and flint glasses.

P.T.O

Q.3. Explain in brief (1 out of 2)

(1 × 10 = 10)

- a.** Derive the formula of equivalent power for combination of two thin lenses.
- b.** With proper diagram explain the rules and image formation by convex and concave mirror.